SOAP web security testing

web services and API's

- software components designed to facilitate communication and data exchange between diff app sys
- using protocol like SOAP or REST
- used for machine to machine

web services VS API's

- web services are a boarder category of tech used to enable machine to machine communication and data exchange
- API is a set of rules and tools that allow developers to access the functionality or data of services app or platform
- API's can interact with humans and machines
- web services provide standardized way for various applications , often platforms and diff programming language to interact with each other
- API's provide access to functionality or data of an app or services for developers to use their own application

Web services implementations

- SOAP (Simple Object Access Protocol)
- exchanging structured info in the implementation of web services by using XML
- JSON-RPC and XML-RPC
    - light weight protocols for remote procedure using JSON or XML
- REST
    - architectural style for designing networked application and it uses HTTP as it's Communication channel

XML RPC

- has been used as precursor to more modern web services like SOAP and REST

- protocol used to encode and decode data in XML format
- it works by sending HTTP requests that call single method
- XML is understandable than JSON
- JSON RPC (each of use , similar of XML -RPC , more human-readable)

SOAP

- set of rules and conventions for structuring messages defining remote procedure call (RPC)
- Known for its strong typing and extensive features set , include security reliability
- known as (WSDL) declaration language
- GET retrieve resource
- PUT change state
- POST new resource
- DELETE delete resource

WSDL language

- web service description language , is an XML-based used to describe the functionality of web service
- WSDL with SOAP to define and document SOAP-Based web services
- WSDL -> 1.1 and 2.0
- Abstract : what service done
- Concrete : adds info about how the web service communicates and where the functionality is offered

WSDL Enumeration

- First access WSDL file
- Identify endpoints
- Identify SOAP services

```
?wsdl,-wsdl
?disco
```

- once getting them start gather info (operations , data , syntax)

Invoking Hidden Methods

-